

Data Centres in Wales

Conditions that should apply to consent for large-scale
data centre development in Wales

Drew Griffiths

This submission does not oppose data centre development in Wales. It argues that development at the scale now proposed should not be consented without binding, testable conditions on how each site is powered, how much water it draws, and how its performance is monitored. It sets out six such conditions, each drafted to be delivered through devolved planning powers.

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Route of submission. This response is submitted by email to contact@senedd.wales because the online submission forms were unavailable at the time of writing. It is offered to the Economy, Energy and Connectivity Committee and to the Local Government, Housing and Planning Committee of the Seventh Senedd, both of which are currently seeking views on their priorities, and to Senedd Research as a response to its briefing of 31 July 2026. I would be grateful if it could be directed accordingly.

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1. Summary

Welsh planning policy does not currently mention data centres. That silence is the problem this submission addresses. Applications of national significance are being determined against a policy framework written for a different kind of industrial building, at a moment when a single proposed site would consume several times the electricity of the entire Welsh data centre estate as it stood in 2024.

I am not opposed to this industry locating in Wales. The economic case is real, and much of the proposed development sits on former industrial land with existing grid and fibre assets. The question is not whether to host it but on what terms, and whether those terms are set before the current pipeline is determined or after.

The submission makes three claims. First, that the scale of the current pipeline is categorically different from the estate Welsh policy was written around. Second, that the obvious conditions, expressed as renewable power and independent generation, do not work as drafted and can be satisfied in ways that make outcomes worse rather than better. Third, that a workable version exists, is enforceable at the point of consent, and sits within devolved planning powers.

2. The scale of what is proposed

The Department for Science, Innovation and Technology estimated Welsh data centre capacity at 154 MW in 2024, the highest outside London in Great Britain. The proposals now in the pipeline are of a different order.

Table 1. Proposed Welsh developments, with indicative annual consumption and the local generation each would imply

Proposed development	Capacity	Indicative annual use	Pen y Cymoedd equivalents
Cardiff (Vantage)	148 MW	1.1 TWh	1.8
Cardiff (Park and Ride site)	150 MW	1.1 TWh	1.9
Newport	200 MW	1.5 TWh	2.5
Anglesey	349 MW	2.6 TWh	4.3
Bridgend	600 MW	4.5 TWh	7.4

Annual use is calculated at 85 per cent average load, which is conservative for facilities of this type. Pen y Cymoedd, the largest onshore wind farm in England and Wales, has 76 turbines, 228 MW of capacity and generates approximately 0.6 TWh a year; the final column expresses each proposal as a multiple of that output. A typical hyperscale facility is 20 to 100 MW, so every proposal in the table sits at or above the top of that range, and one sits far beyond it.

The aggregate position is consistent with these figures. The UK Government has said the South Wales AI Growth Zone alone has the potential to harness over 1 GW by the early 2030s, which is approximately 13 per cent of the 7.9 GW of total Welsh electricity generating capacity recorded in 2024. The Department for Energy Security and Net Zero found that data centre electricity consumption across south-west England and Wales doubled between 2020 and 2024. The House of Commons Library estimates that data centres could account for between 0.9 and 3.4 per cent of UK greenhouse gas emissions between 2025 and 2035, largely because of non-renewable electricity supply. These loads have to be reconciled with Welsh carbon budgets, and at present nothing in Welsh planning policy requires that reconciliation to be demonstrated at application.

Grid capacity is already binding. Ofgem has stated that there are too many data centres in the connection queue. As of May 2026 the National Energy System Operator held in the region of 173 data centre connection requests representing over 100 GW. This matters for the conditions proposed below, because a long queue is precisely what pushes developers towards generating their own power on site, by whatever means is quickest.

3. Why a renewable requirement must be written as matching

The certificate problem. An operator can describe a facility as 100 per cent renewable while drawing from a gas-heavy grid on every still winter night. The claim is normally satisfied by purchasing certificates equal to annual consumption, so a shortfall in one hour is netted against a surplus in another. Annual accounting of this kind is compatible with a facility whose actual hour-by-hour demand is met substantially by unabated gas. A condition requiring renewable supply, without specifying the accounting period, is therefore satisfied on paper and unenforceable in substance.

The local generation problem. Requiring developers to build their own renewable generation locally sounds stricter, and in one respect it is, but the arithmetic in Table 1 shows the limit. The Bridgend proposal alone would require the annual output of roughly seven Pen y Cymoedds. Solar in Wales, at roughly a tenth capacity factor, is materially worse per hectare. No developer will build generation at that scale as a condition of consent. In practice a requirement phrased this way produces a token on-site array alongside continued grid draw, or, worse, on site fossil generation.

The behind-the-meter problem. That second outcome is not hypothetical. UK operators have been reported as planning to burn gas on site to generate their own electricity, precisely because connection queues are long. A condition requiring independence from the grid, without a fuel restriction, would actively reward that choice. It would deliver a facility that is independently powered, locally generating, and materially worse for Welsh carbon budgets than one connected to the grid.

The conditions in section 4 are drafted to close all three routes. The operative change is to require hourly matching against additional Welsh generation, and to prohibit unabated fossil generation on site, rather than to require ownership or independence as such.

4. Recommended conditions

Condition 1: explicit treatment in national planning policy

Data centres should be named in Planning Policy Wales and in the National Development Framework, with a capacity threshold above which a proposal is treated as a Development of National Significance. England and Scotland already name data centres in their planning frameworks. Until Wales does, local planning authorities have no clear basis on which to weigh these applications, and no settled basis on which to refuse one. This condition is the prerequisite for the five that follow, because it is what gives them a home in policy.

Condition 2: hourly matched, additional renewable supply

- Consumption must be matched to renewable generation on an hourly basis, not netted annually. This is the single provision that distinguishes genuine renewable operation from certificate purchasing.
- Matched generation must be newly commissioned capacity in Wales, commissioned for the development. Reassigning output from existing Welsh wind by contract should not qualify, since it displaces other consumers rather than adding supply.
- Where hourly matching cannot be met from generation, it may be met from co-located or contracted storage, sized against the modelled shortfall and secured at consent.
- Grid supply should be permitted as a genuine fallback, with an annual cap on unmatched hours declared at application and enforced by condition.

Condition 3: no unabated fossil generation on site

Backup generation should be restricted to emergency use, with run hours capped, metered and reported annually. Any proposal for on-site gas turbines or reciprocating engines intended for routine or primary supply should be refused. Without this condition, Condition 2 can be satisfied by burning gas behind the meter, and the policy produces the opposite of its intent.

Condition 4: water use and water quality

- Closed-loop or air cooling should be the default. Evaporative cooling drawing on potable supply should require specific justification at application.
- A hard cap on potable abstraction should be set at consent, expressed in cubic metres per year, alongside a water usage effectiveness target.

- Rainwater capture and reuse should be required as a supplementary source. It should not be accepted as the primary mitigation, because harvesting at any realistic roof area cannot meet cooling demand at these loads.
- Discharge should be permitted and monitored by Natural Resources Wales, covering volume, temperature and cooling water chemistry. Blowdown from cooling systems carries biocides, corrosion inhibitors and scale inhibitors, and these should be specified and monitored rather than assumed benign.

The direction of travel makes this more pressing rather than less. Liquid cooling is more efficient than air cooling and is used for the high-density equipment that AI workloads require, so its use and scale in Wales is likely to increase as these developments proceed.

Condition 5: monitoring, publication and enforcement

- Mandatory metering of electricity consumption, power usage effectiveness, water usage effectiveness, potable and non-potable draw, backup generator run hours and discharge quality.
- Quarterly publication of those figures in a standard format, accessible to the host community.
- A defined enforcement pathway triggered by breach of a declared threshold. A condition that is not enforceable is not a condition, and self-reported annual sustainability statements are not monitoring.
- Boundary noise limits set at consent and monitored continuously. Noise is a documented impact of these facilities with health and environmental consequences, and it is among the concerns already raised by communities in Wales.

Condition 6: securing local benefit

The UK Government projects around 3,400 jobs in the North Wales AI Growth Zone and at least 5,000 in South Wales, figures which include temporary construction roles. Once operational, data centres are highly automated and employ a permanent workforce smaller than their land and resource use would suggest. Analysis published by Action to Protect Rural Scotland in March 2026 argues that UK Government job estimates are not accurate, and evidence from the United States indicates that data centres do create local jobs, but fewer than industry advocates claim. Consent should therefore secure:

- A stated floor for permanent operational posts, with annual reporting against it.
- A contribution to grid reinforcement costs, so that network upgrades driven by these loads are not socialised onto household bills.
- A community benefit fund on the model already established for Welsh onshore wind, with duration matched to the operational life of the site.
- Biodiversity net benefit consistent with Welsh statutory biodiversity targets.

Table 2. The six conditions, and what each is designed to prevent

Condition	Failure it is drafted to prevent
1. Named in national planning policy	Applications of national significance determined with no policy basis for refusal
2. Hourly matched, additional supply	Annual certificate netting presented as renewable operation
3. No unabated fossil on site	Behind the meter gas adopted to bypass the connection queue
4. Water caps, closed-loop, discharge permitting	Potable abstraction and untested discharge chemistry treated as a detail
5. Metering, publication, enforcement	Conditions that exist on paper and are never tested
6. Job floors, grid contribution, community fund	Wales hosting the infrastructure without securing the benefit

5. Devolved competence: why these conditions run through planning

Electricity market regulation, licensing and network charging are reserved to the UK Government and to Ofgem. Wales cannot impose supply conditions on operators through energy legislation, and a submission that asked it to would be asking for something the Senedd cannot deliver.

Every condition above is therefore framed to be delivered through devolved planning powers: national planning policy, development plan policy, planning conditions, and planning obligations. That is a deliberate design choice rather than a limitation reluctantly accepted. It means the package can be implemented by Welsh Ministers and local planning authorities without waiting for reserved powers to change, and without intergovernmental agreement.

Two consequences follow. Conditions must be attached to consent rather than negotiated afterwards, because planning conditions are enforceable and voluntary commitments are not. And the threshold question in Condition 1 matters disproportionately, because it determines whether the largest proposals are handled locally or as Developments of National Significance, and therefore whether these conditions are applied consistently across Wales or vary between authorities facing very different levels of developer pressure.

6. What this submission asks for

The Cabinet Minister has said that Wales should not merely host this infrastructure but should get the benefit of it. I agree, and this submission is an argument about the mechanism by which that benefit is secured. Benefit that is not written into consent is not secured at all.

Specifically, I ask that a committee take up data centre consenting as an early inquiry topic in the Seventh Senedd, and that the inquiry address:

1. Whether Planning Policy Wales and the National Development Framework should name data centres, and at what capacity threshold a proposal becomes a Development of National Significance.
2. Whether hourly matching against additional Welsh generation can be secured by planning condition, and what evidence an applicant should be required to produce to demonstrate it.

3. What monitoring and publication regime should attach to consent, and which body should hold and enforce it.
4. What the cumulative electricity and water demand of the current pipeline implies for Welsh carbon budgets and for water resource management plans, assessed together rather than application by application.

The pipeline is being determined now. Policy written after these consents are granted will apply to a much smaller share of the industry than policy written before, and conditions cannot be retrofitted to a permission already granted. That timing, rather than any objection to the industry itself, is the reason for making this submission at this point.

Sources

1. Senedd Research, *Data centres in Wales: How will Wales manage a rapidly growing industry?*, 31 July 2026, and the accompanying research briefing.
2. Department for Science, Innovation and Technology, *Estimate of data centre capacity, Great Britain, 2024*.
3. Department for Energy Security and Net Zero, *Energy Trends, June 2026: data centre electricity consumption in Great Britain, 2020 to 2024*.
4. House of Commons Library, research briefing CBP-10315, data centres.
5. Ofgem, *Demand Connections Call for Input*, February 2026.
6. Welsh Government, *Energy generation and use in Wales: first combined report*.
7. Parliamentary Office of Science and Technology, POSTnote 0762, data centre energy and water use.
8. Vattenfall and Natural Resources Wales, published operating data for the Pen y Cymoedd wind energy project.
9. Action to Protect Rural Scotland, jobs and data centres, March 2026.
10. Brookings Institution, new evidence on data center employment effects.
11. UK Government, AI Growth Zone announcements for North Wales and South Wales.

Prepared as an individual written submission. Capacity and consumption figures are drawn from the published sources listed above; the annual consumption and generation equivalents in Table 1 are the author's own calculations from those figures, at a stated load assumption, and are indicative rather than an engineering assessment of any specific site.